Anirudha Kemtur

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AKemtur

Education

2020–2023 **Masters in Computer Science**, *MILA - Quebec AI Institute - University of Montreal*, Montreal, Canada, GPA 3.8/4.3, Advisors: Dr Karim Jerbi and Dr Pierre Bellec
AI-based modeling of brain and behavior: Combining neuroimaging, imitation learning and Video games

2015–2020 **B.E.(Hons.) in Computer Science and M.Sc.(Hons.) in Economics**, *Birla Institute of Technology and Science (BITS) Pilani*, Pilani Campus, India, GPA 8.76/10, Advisor: Dr. Falk Lieder
Improving Human Decision-Making using Metalevel-RL and Bayesian Inference

Experience

May'26–Present **AI Engineer**, *Janelia Research Campus*, Howard Hughes Medical Institute (HHMI), Ashburn,USA
Dr. Rob Johnson and Dr. Josh Dudman
- Training a MuJoCo-based digital rat using imitation learning to replicate real rat motion and behavior for biomechanical analysis.
- Computational modeling of rodent behavioral experiments.

Skills: *Deep Reinforcement Learning, Computational Neuroscience*

May'23–May'26 **Machine learning SCA**, *Janelia Research Campus*, Howard Hughes Medical Institute (HHMI), Ashburn,USA
Dr. Rob Johnson
- Building hierarchical controllers of rodent locomotion.

Skills: *Deep Reinforcement Learning, Biomechanics*

September'20–April'23 **Graduate Researcher**, *MILA - Quebec AI Institute*, Université de Montréal, Montreal,Canada
Dr. Karim Jerbi and Dr. Pierre Bellec
- Worked on Courtois Neuromod Project where human brain and gameplay data is collected while participants play videogame inside fMRI and EEG brain scanner.
- Trained Imitation learning networks to play the videogame while exhibiting gameplay style similar to subjects and developed models to encode fMRI brain data from activations of the imitation learning network.

[Article] **Skills:** *Deep Reinforcement Learning, Neuroimaging, Imitation learning, Videogames*

February'20–August'20 **Machine Learning Intern**, *Computational and Cognitive Neuroscience Lab*, Université de Montréal, Montreal,Canada
Dr. Karim Jerbi
- Developed methods to do domain transfer learning between sleep and anesthesia human brain data using domain adversarial neural networks.
- Developed a pipeline using guided backpropagation to visualize brain features extracted by the model.

[Article] **Skills:** *Transfer learning, Domain adaptation, Feature visualization*

September'19–December'19 **Undergraduate Thesis**, Max Planck Institute for Intelligent Systems, Tübingen, Germany
Dr. Falk Lieder
- Trained deep neural networks to discover strategies robust to Human biases using Meta-level Reinforcement learning and Bayesian Inference.
- Discovered strategies were taught to humans and was shown to improve their decision-making.

[Article] **Skills:** *Sequence modelling, Meta-RL, Bayesian Inference*

May'19–August'19 **MITACS Globalink Intern**, *Computational and Cognitive Neuroscience Lab*, Université de Montréal, Montreal, Canada
Dr. Karim Jerbi
- Study of EEG sleep data using Convolutional neural networks.

[Article] **Skills:** *: Computer vision, Medical Imaging, Neuro-AI*

Publications and Poster presentations

July'26 **Behavioral imitation with artificial neural networks leads to personalized models of brain dynamics during videogame play**, *Kemtur et al. 2026*, Imaging Neuroscience
[Article]

Nov'25 **Rats learn and use intuitive physics knowledge to solve fetch tasks**, *Yan et al. 2025*, Society for Neuroscience
[Article]

August'23 **Class imbalance should not throw you off balance: Choosing the right classifiers and performance metrics for brain decoding with imbalanced data**, *Thölke et al. 2023*, NeuroImage
[Article]

October'22 **Leveraging Machine Learning to Automatically Derive Robust Decision Strategies from Imperfect Knowledge of the Real World**, *Mehta et al. 2022*, Journal of Computational Brain and Behavior
[Article]

June'22 **Comparing sleep and anesthetic-unconsciousness using Domain adversarial neural networks and EEG**, *Kemtur and Ghosh et al. 2022*, OHBM 2022
[Article]

December'20 **Improving Human Decision-Making using Metalevel-RL and Bayesian Inference**, *Kemtur and Jain et al. 2020*, NeurIPS RWRL 2020
[Article]

June'20 **Leveraging Machine Learning to Automatically Derive Robust Planning Strategies from Biased Models of the Environment**, *Kemtur and Jain et al. 2020*, Proceedings of the 42nd Annual Conference of the Cognitive Science Society(p. 2405-2411), CogSci 2020
[Article]

Selected Awards and Honors

December'22 **Best poster award-MAIN-Montreal**, Award of 250 CAD

January'21 **Microsoft AI grant**, Financial award of 4000 CAD

September'20 **Mitacs graduate fellowship award**, Financial award of 15000 CAD to pursue masters degree

September'20 **University of montreal matching award**, Graduate tuition fee exemption scholarship of 15000 CAD

February'20 **Mitacs research fellowship**, Financial award of 6000 CAD to participate as research exchange student in Canada for 4 months

August'19 **BITS-Pilani Thesis scholarship**, Financial award of 100000 INR to pursue off-campus bachelor thesis in Germany

April'19 **Mitacs Globalink Scholarship**, Financial award of 5000 CAD to participate as research intern in Canada for 3 months

August'15-August'18 **ACM - Best Student Chapter(India) Award for 3 consecutive years**, Key member of BITS-Pilani Association for Computing Machinery Student chapter.

June'15 **Jee Mains Exam - All India Rank-3320**, 1.3 Million Students appeared for the exam

Skills

Expertise Reinforcement learning , Imitation learning, Computer vision, Biomechanics, Medical imaging,

Languages **Proficient**:: Python **Familiar**: Java, C, Verilog

Libraries **Proficient**:: PyTorch, Matplotlib, Sklearn, OpenAI gym **Familiar**: Mujoco, Unity

Invited talks

Invited talks **Mitacs-MTL event, Montreal** - Invited as a panelist to share my research journey and discuss AI opportunities in Montreal with over 300 international mitacs interns.(2022)

Department of Education - Govt of India, Delhi - Invited to share feedback and recommendations to improve India-Canada research collaborations.(2020)